Status	Finished
Started	Thursday, 15 October 2026, 8:51 PM
Completed	Thursday, 15 October 2026, 8:54 PM
Duration	3 mins 7 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

a. Hash search
b. Linear search
c. Sequential search
d. Binary search ✓

Question **2**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

a. By starting from the first element
b. By using a hash function
c. By dividing the list length by two ✓
d. By comparing each element sequentially

Question **3**

Correct

Mark 1.00 out of 1.00

What is the first step in binary search?

a. Compare the target element with the middle element in the list ✓
b. Compare the target element with the first element in the list
c. Sort the list
d. Divide the list into two equal parts

Question **4**

Correct

Mark 1.00 out of 1.00

_____ search takes a sorted/ordered list and divides it in the middle.

- a. Binary ✓
- b. Hash
- c. Linear
- d. Both (1) & (3)

Question **5**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

- a. 500
- b. 10 ✓
- c. 100
- d. 1000

Question **6**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

- a. n/2
- b. n ✓
- c. n-1
- d. log n

Question **7**

Correct

Mark 1.00 out of 1.00

Which method of searching involves sequentially comparing each element until a match is found?

- a. Binary search
- b. Hashing
- c. Jump search
- d. Linear search ✓

Question **8**

Correct

Mark 1.00 out of 1.00

What is the key characteristic of binary search?

a. It works on unsorted lists
b. It always starts from the beginning of the list
c. It can be applied only if the list is sorted ✓
d. It compares elements sequentially

Question **9**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

a. Must use a sorted array
b. Binary search algorithm is not efficient when the data elements more than 1500 ✓
c. There must be a mechanism to access middle element directly
d. Requirement of sorted array is expensive when a lot of insertion and deletions are needed

Question **10**

Incorrect

Mark 0.00 out of 1.00

What happens in a binary search if the list has an even number of elements?

a. The higher middle element is chosen as the middle element
b. The search stops
c. The lower middle element is chosen as the middle element ✗
d. The middle element is chosen randomly

Question **11**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search requires the list to be sorted.
b. Linear search is more efficient than binary search on large lists.
c. Linear search can be applied to both sorted and unsorted lists. ✓
d. Linear search divides the list into halves.

Question **12**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

a. When the item is not the array at all

b. When the item is somewhere in the middle of the array ✓

c. When the item is the last element in the array

d. Item is the last element in the array or item is not there at all

Question **13**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

a. The list is sorted

b. The search continues in the left or right sublist ✓

c. The search continues from the beginning

d. The search stops

Question **14**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

a. In the right sublist

b. In the left sublist ✓

c. At the beginning of the list

d. In the entire list

Question **15**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

a. When the list contains duplicate elements

b. When the element is found in the list ✓

c. When the list is sorted

d. When the search algorithm finishes